Question1: Multiple Choice Question (10) Minutes [4] / 1x5 = 5 Marks

YOU MUST USE THE PROVIDED TABLE BELOW. USE CAPITAL LETTERS (A,B,C and D)

1	2	3	4	5
B	A	~~B~~ C	D	C

1. What is an algorithm:
   A. A flowchart
   B. Step by step instructions used to solve a problem
   C. A decision
   D. A pseudocode

2. How is time complexity measured?
   A. By counting the number of primitive operations performed by the algorithm on a given input size.
   B. By counting the number of algorithms in an algorithm.
   C. By counting the size of data input to the algorithm.
   D. None of the above

3. What is the time complexity of the following code:

```
int i, j, k = 0;
  for (i = n / 2; i <= n; i++) { logn } n log n
    for (j = 2; j <= n; j = j * 2) { n }
      k = k + n / 2; }}
```

   A. $O(n)$
   B. $O(n^2)$
   ✓ C. $O(n \log n)$
   D. $O(n^2 \log n)$

4. Find the slowest time:
   A. $O(n)$
   B. $O(n^2)$
   C. $O(2^n)$
   D. $O(n!)$

5. The complexity of the below recurrence $T(n) = \begin{cases} 3T(n-1), & n > 0 \\ 1, & n = 0 \end{cases}$ is:
   A. $O(n^3)$
   B. $O(n^n)$
   C. $O(3^n)$
   D. $O(n \log n)$

Question 2: **(20) Minutes [4.5] / 5 Marks**

1. Find the relation between the functions: **[2 marks]**

a. $f(n) = 2^{\log n}$ and $g(n) = n^{\sqrt{n}}$

$f(n) \leq g(n)$ $f(n) = O(g(n))$ 2

b. $f(n) = 2n$ and $g(n) = 3n$

$f(n) \leq g(n)$ $f(n) = O(g(n))$

c. $f(n) = 2^{n+1}$ and $g(n) = 2^n$

$f(n) \geq g(n)$ $f(n) = \Omega(g(n))$

d. $f(n) = 2^{(2n)}$ and $g(n) = 2^n$

$f(n) \geq g(n)$ $f(n) = \Omega(g(n))$

2. Use the *iteration method* to give the tight upper bound (big-O) to the following recurrence: **[3 marks]**

$$T(n) = \begin{cases} T\left(\frac{n}{3}\right) + n, & n > 1 \\ 1, & n = 1 \end{cases}$$

$T(n) = T\left(\frac{n}{3}\right) + n$

$T\left(\frac{n}{3}\right) = T\left(\frac{n}{3^2}\right) + \frac{n}{3}$

$T(n) = \left(T\left(\frac{n}{3^2}\right) + \frac{n}{3}\right) + n$

$T\left(\frac{n}{9}\right) = T\left(\frac{n}{3^3}\right) + \frac{n}{3^2}$ 2.5

$T(n) = \left(T\left(\frac{n}{3^3}\right) + \frac{n}{3^2}\right) + \frac{n}{3} + n$

$T(n) = \left(T\left(\frac{n}{3^4}\right) + \frac{n}{3^3}\right) + \frac{n}{3^2} + \frac{n}{3} + n$

$T(n) = T\left(\frac{n}{3^k}\right) + \frac{n}{3^{k-1}} + \frac{n}{3^{k-2}} + \cdots + \frac{n}{3^0}$

$k = \log_3 n$

$T(n) = \frac{n}{3^k} = \frac{n}{3^{\log n}}$

$T(n) = T(1) + n$

$T(n) = T(1) + \frac{n}{3^k}$ $T(n) = (1) + \frac{n}{3^{\log n}}$ $O(n)$